Question 1 Planning (15 marks)

Defining the problem (2 marks)

P λ is the independent variable, or vary λ. [1]

P V is the dependent variable, or measure V. [1]

Methods of data collection (4 marks)

M Circuit diagram showing d.c. power supply in series with diode (correct symbol needed) and method to measure potential difference across diode. Circuit must be correct. [1]

M Instrument to change p.d. across LED e.g. variable power supply/potential divider/variable resistor. [1]

M Record wavelength of light of LED from data sheet or use Young's slits/diffraction grating. [1]

M (Slowly) increase potential difference across LED until LED (just) emits light (or reverse procedure). [1]

Method of analysis (3 marks)

A Plot a graph of lg V against lg λ (allow natural logs). Allow lg λ against lg V. [1]

A n = gradient [1]

A $k = 10^{y\text{-intercept}}$ [1]

Additional detail (6 marks)

Relevant points might include: [6]

1 Use of a protective resistor (can be shown on the diagram).

2 Polarity of LED correct in circuit diagram.

3 Instrument to determine when LED just lights e.g. light meter/detector, LDR.

4 Method to use light detector/LDR to determine point at which LED emits light.

5 Expression that gives λ (symbols need to defined) from experimental determination of wavelength of light, e.g. Young's slits/diffraction grating.

6 Perform experiment in a dark room/LED in tube.

7 Relationship is valid if graph is a straight line.

8 $\lg V = n \lg \lambda + \lg k$

9 Repeat V and average for the same λ or LED.

Do not allow vague computer methods.

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